

QVerse — Week 9: Quantum Protocols — Teleportation and Superdense Coding

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

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How this booklet fits the course

This theory booklet is designed to be read alongside `09_teleportation_superdense_coding.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 9; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Explain what quantum teleportation transfers.
- Implement a coherent/deferred-measurement teleportation circuit.
- Implement superdense coding.
- Identify the role of entanglement and classical information.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. Why quantum communication protocols matter

Teleportation and superdense coding are ideal examples of how quantum resources change information-processing tasks without violating ordinary causality.

Both protocols use a pre-shared entangled pair, but they solve different problems:

- **Quantum teleportation:** transfer an unknown qubit state using one shared Bell pair plus two classical bits.
- **Superdense coding:** transmit two classical bits by physically sending one qubit, provided one Bell pair was shared beforehand.

Neither protocol creates free communication. The accounting of entanglement, transmitted qubits, and classical messages is essential.

2. Teleportation: the task

Alice possesses an unknown message qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

Alice and Bob also share the Bell pair

$$|\Phi^+\rangle_{AB} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

The initial three-qubit state is

$$|\psi\rangle_M \otimes |\Phi^+\rangle_{AB}.$$

The goal is for Bob's qubit to end in state $|\psi\rangle$ without Alice physically sending the message qubit itself.

3. Resources required for teleportation

Textbook teleportation consumes:

- one unknown input qubit;
- one Bell pair shared in advance;
- two classical bits sent from Alice to Bob.

After the protocol, Bob holds the original quantum state. Alice does not retain an independent copy.

This is fully consistent with the no-cloning theorem.

4. Bell-basis processing

Alice performs a CNOT from the message qubit to her half of the Bell pair, followed by a Hadamard on the message qubit.

After those gates, the three-qubit state can be rearranged as

$$\frac{1}{2} \left[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\beta|0\rangle + \alpha|1\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(-\beta|0\rangle + \alpha|1\rangle) \right],$$

where the first two kets represent Alice's two measured qubits and the final ket is Bob's qubit.

This expression shows why two classical correction bits are enough. Depending on Alice's result, Bob's state is one of four versions of $|\psi\rangle$ related by X and Z .

5. Classical correction rule

Using one common convention, Alice obtains two classical bits m_1, m_2 . Bob applies

$$X^{m_2} Z^{m_1}$$

to recover $|\psi\rangle$.

The exact visual ordering of the two bits depends on circuit and software conventions, so students should label qubits and classical registers explicitly rather than memorize a displayed string.

The conceptual rule is stable:

- one measurement bit determines whether an X correction is required;
- the other determines whether a Z correction is required.

6. Why teleportation does not send information faster than light

Before receiving Alice's classical bits, Bob does not know which correction to apply.

Locally, Bob's qubit is described by a maximally mixed state. He cannot extract $|\psi\rangle$ from it by himself.

Therefore the protocol cannot be completed faster than the classical message can travel.

Entanglement supplies nonclassical correlations; classical communication supplies the information needed to make the transferred state usable.

7. Deferred measurement principle

The notebook uses a coherent or deferred-measurement form of teleportation.

The **deferred measurement principle** says that, in many circuits, a measurement followed by classically controlled gates can be replaced by equivalent coherent controlled operations and a later measurement, provided the circuit is transformed correctly.

This is useful because simulators often make it convenient to inspect the final state before measurement.

Conceptually, the protocol is still teleportation: the coherent controls stand in for measurement outcomes and classical feed-forward.

8. Fidelity

To test teleportation, compare the target state $|\psi\rangle$ with Bob's output state $|\phi\rangle$.

For pure states, the state fidelity is

$$F = |\langle\psi|\phi\rangle|^2.$$

Ideal teleportation should give

$$F = 1$$

up to numerical precision and global phase.

On real hardware, noise can reduce the observed fidelity.

Fidelity is a better validation metric than testing only $|0\rangle$ and $|1\rangle$, because a correct teleportation circuit must preserve arbitrary amplitude and phase information.

9. Testing a family of input states

A convenient family is

$$|\psi(\theta, \phi)\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle.$$

By varying θ and ϕ , one can test many points on the Bloch sphere.

Testing only computational-basis states would not catch errors that destroy relative phase.

10. Superdense coding: the task

Alice and Bob again start with a shared Bell pair.

Alice wishes to send one of four classical messages:

$$00, \quad 01, \quad 10, \quad 11.$$

She encodes the message by applying one of four local operations to her qubit:

$$I, \quad X, \quad Z, \quad XZ$$

or an equivalent convention.

She then physically sends her qubit to Bob.

Bob now possesses both qubits and performs a Bell-basis decoding circuit. The final computational-basis measurement reveals two classical bits.

11. Why superdense coding is not “two bits from nothing”

The protocol uses a resource that was established earlier: shared entanglement.

Its accounting is:

- entanglement distributed in advance;
- one qubit transmitted during the messaging stage;
- two classical bits recovered.

The protocol does not violate information-theoretic limits because the pre-shared entanglement is part of the resource budget.

12. Bell-basis decoding

A typical decoder uses

1. CNOT;
2. Hadamard;
3. computational-basis measurement.

This is the inverse of the common Bell-state preparation circuit.

Thus the four orthogonal Bell states are mapped to four distinct computational-basis bit strings.

This is an important general pattern: a basis that is difficult to measure directly can often be transformed into the computational basis before measurement.

13. Teleportation and superdense coding compared

Protocol	Shared resource	Communication during protocol	Output
Teleportation	1 Bell pair	2 classical bits	1 unknown qubit state transferred
Superdense coding	1 Bell pair	1 qubit	2 classical bits transferred

The two protocols are closely related because both exploit Bell-basis structure and pre-shared entanglement.

14. Common misconceptions

“Teleportation moves matter”

It transfers a quantum state, not a physical object.

“Teleportation copies the state”

The unknown state is transferred; no independent copy remains at Alice’s side.

“Entanglement alone is enough”

No. Teleportation requires two classical bits.

“Superdense coding lets one qubit permanently store two classical bits”

The protocol relies on a joint two-qubit entangled system and pre-shared resources.

15. What to know before the notebook

You should be able to explain:

- the resource accounting of both protocols;
- the Bell-pair role;
- why teleportation needs X and Z corrections;
- why classical communication prevents faster-than-light signalling;
- why fidelity is a useful teleportation metric;
- how Bell-basis decoding enables superdense coding.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.